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Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®
JAIN COLLEGE OF ENGINEERING, BELAGAVI



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

PRE-FINAL YEAR
(2025 – 2026)

MINI-PROJECT REPORT

On

"EFFECTIVE AQUAPONIC FARMING "

PROJECT GUIDE

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CERTIFICATE

This is to certify that the **Mini-Project Work** entitled “**EFFECTIVE AQUAPONIC FARMING**” carried out by **Rasika Bhaskar (2JI24EC408), Sakshi Tarale (2JI24EC411), Vishal Nalkalkar(2JI23EC179) and Vivek NallaVade (2JI23EC183)** are bonafide students of **Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi**, in partial fulfilment for the award of **Bachelor of Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2024-2025**. It is certified that all corrections/suggestions indicated for project assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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VISION

To achieve excellence in education and research for developing globally competent , ethically sound electronic and communication engineers

MISSION

1. To provide a conducive environment for learning through structured student centric, teaching learning process
2. To nurture needs of society by infusing scientific temper in students and to grow as a centre of excellence with efficient industry-institute interaction
3. To inculcate self learning skills, entrepreneurial ability and professional ethics.

Program Educational Objectives (PEO's)

Graduates will be able to

- Contemplate real-time social problems and deliver efficient solutions
- Lead and succeed in professional careers
- Contribute through research and entrepreneurship

Program Specific Outcomes (PSO's)

Graduates in the UG program in Electronics and communication engineering will be able to

- Design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs.
- Apply the knowledge in various domains of IoT, real time systems, communication systems, VLSI and embedded systems, image and signal processing using hardware and software tools.



PROGRAM OUTCOME'S (PO'S)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the **engineering and management principles and apply these to one's own work, as a member and leader** in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



Course Outcomes (CO's)

Course Outcome	Course Outcome Statement	Bloom's Level
BEC586.01	Design and develop an electronic system using modern tools by following professional ethics.	L3,L4
BEC586.02	Develop managerial, teamwork and communication skills.	L3
BEC586.03	Understand the design factors to be considered that impact society.	L2,L3

Bloom's Cognitive Levels

L1: Remembering; L2: Understanding; L3: Applying; L4: Analyzing; L5: Evaluating; L6: Creating

STRENGTH OF CO MAPPING TO PO/PSOS WITH JUSTIFICATION:

1. Slight (Low) 2. Moderate (Medium) 3. Substantial (High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	1	3	-	-	2	-	-	-	2	2	2
CO2	-	-	-	-	-	-	-	-	3	3	2	-	-	-
CO3	-	-	-	-	-	2	2	-	-	-	-	-	-	-

CO – PO - PSO	Justification
CO1 → PO1 (3) CO1 → PO2 (3) CO1 → PO3 (3) CO1 → PO4 (1) CO1 → PO5 (3) CO1 → PO8 (2) CO1 → PO12 (2) CO1 → PSO1(2) CO1 → PSO2(2)	- Students apply the electronics and communication engineering skills and design products to solve complex engineering problems. - Students gain the knowledge of modern tools and skill usage. - Students can use research-based knowledge and methods including design of experiments, analysis and interpretation to provide valid conclusions. - Students can work ethically and professionally in the industry. - Students can design their projects in analog or digital fields. - Students can choose any technology to build their projects.
CO2 → PO9 (3) CO2 → PO10 (3) CO2 → PO11 (2)	- Students can effectively act as an individual, and as a member or leader and work in a team. - Students can comprehend and write effective reports and make documentation with effective presentation. - Learn to management and financial skills required for the execution of project.
CO3 → PO6 (3) CO3 → PO7 (3)	- Students can able to understand the impact of professional engineering solutions in societal and environmental contexts. - Students can be able to apply reasoning of contextual knowledge to assess societal, health and safety issues.

ACKNOWLEDGEMENT

Although a single sentence hardly suffices, we would like to thank almighty God for blessing us with his grace and taking our endeavour to a successful culmination.

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We extend our sense of gratitude to **Prof.V.R.Bagali , Head, Dept. of E&CE, JCE, Belagavi**, for extending support and cooperation which helped us in completion of the project work.

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ABSTRACT

This technical project begins by introducing the concept of aquaponics, including a brief history of its development and its place within the larger category of soilless culture and modern agriculture. It discusses the main theoretical concepts of aquaponics, including the nitrogen cycle and the nitrification process, the role of bacteria, and the concept of balancing an aquaponic unit. These project works on the three groups of living organisms (bacteria, plants and fish) that make up the aquaponic ecosystem. The most prevailing issues of the modern world are food and water crises. It is neither possible to consume the pesticide affected food nor grow one's own plants, due to scarcity of water and land. Under such conditions, there arises a need for a portable agricultural system which uses less water, space and is purely organic. One such solution is a small scale aquaponic system. This system is made by introducing an automation there by there is no need for setting aside extra time for system care. This project has used the data acquired from an existing aquaponic system to design and implement an effective small scale sustainable aquaponic system. This can lead to cost effective, sustainable ways of organic farming independent of the need for comparable land space requirement.

Dissolved chemical element is important for your fishes. Your lake establishes adequate levels of dissolved chemical element through correct aeration. Poor aeration is often one of the fastest and most silent killers of your fish. Though you've got had a long time lake for years, your fish would require a lot of chemical element as they grow larger. Night long your fish might die with fully no notice or symptoms. For your fish to thrive, it's counseled that your lake contains 7-9 mg/L of dissolved chemical element.

Water varies from place to position, reckoning on the condition of the supply water from that it's drawn and also the treatment it receives. The standard methodology of testing cloudiness, pH scale & Temperature is to gather samples manually and so send them to laboratory for analysis. However, it's been unable to fulfill the strain of water quality observance these days. Thus a collection of observance of cloudiness, pH scale & Temperature of Water quality has been developed. The system consists of cloudiness, pH scale & Temperature detector of water quality testing, observance accessories like sensors and controllers.

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CHAPTER 1

INTRODUCTION

The project contains the methodology to build a aquaponic system suitable for different economic strata of the society especially focusing on the urban population where there is evident space and time constraints. This method contributes to one aspect of sustainable household development. In a small scale Aquaponic system, organic vegetables are cultivated in a limited space by recirculating water from a fish tank, rich in nutrients which are essential for the plant growth. Out of all the available water resources on planet Earth, 2.5% is freshwater resource. In this 2.5%, only 0.3% is the readily available freshwater resource accessible to humans. 70% of this limited amount of freshwater available is used for agriculture. Water scarcity already affects almost every continent and more than 40 percent of the people on our planet. By 2025, 1.8 billion people will be living in countries or regions with absolute water scarcity and two- thirds of the world's population would be living under water stressed conditions. In 2030, 47% of world population will be living in areas of high water stress. Most population growth will occur in developing countries, mainly in regions that are already experiencing water stress and in areas with limited access to safe drinking water and adequate sanitation facilities. Most of our food requires 100s of liters of water for production and adequate of per crop area for cultivation. The daily drinking water requirement per person is 2-4 liters, but it takes 2000 to 5000 liters of water to produce one person's daily food. In such a situation, a method like aquaponics which is the combination of hydroponics and aquaculture, can contribute effectively to the problem by lowering the amount of water usage for cultivation by 80% and also 75% of the area requirement. This project is mainly implemented on agriculture and production using natural as well as technical ideas. It is an innovative technique which is brought into play by combination of hydroponics and aquaculture. It can help to reduce oil usage. As the popularity in Aquaponics will increase, the need for chemical nutrients which are derived from oil will decrease, thus reducing oil usage. Aquaponics will reduce fresh drinking water consumption levels by being the most efficient use of water for growing vegetables. It has the most efficient use of nutrients out of all forms of agriculture.

1.1 MOTIVATION

Aquaponic farming offers a sustainable solution to modern agricultural challenges such as water scarcity, soil degradation, and rising food demand. Its ability to combine fish cultivation with plant production makes it an eco-friendly alternative to traditional farming. Many communities still lack access to fresh, chemical-free produce, motivating the need for efficient systems like aquaponics. Developing a reliable and easy-to-manage aquaponic setup can empower farmers with new opportunities for income and food security. The project is driven by the desire to promote sustainable practices that reduce environmental impact. It also aims to inspire innovative farming methods that can be adopted in both rural and urban settings. Ultimately, the motivation is to create a system that is practical, affordable, and beneficial for future generations.

1.2 PROBLEM STATEMENT

Effective aquaponic farming faces significant challenges that hinder its widespread adoption. Many farmers struggle with balancing water quality, nutrient levels, and fish-plant symbiosis, which affects overall productivity. Limited technical knowledge and inadequate monitoring systems often lead to system failures. High initial setup costs and inconsistent maintenance practices further reduce efficiency. Therefore, identifying strategies to optimize system management is essential to improve the reliability and scalability of aquaponic farming.

1.3 OBJECTIVE OF PROJECT

The objective of this project is to develop an efficient aquaponic farming system that enhances plant and fish productivity. It aims to optimize water quality, nutrient flow, and environmental conditions for sustainable growth. The project seeks to introduce simple monitoring methods to reduce technical errors. Ultimately, it strives to create a cost-effective, scalable model suitable for both small and large-scale farmers.

1.4 DESIGN APPROACH

1.4.1 Firmware Design & Development

- Requirement Gathering
- Software Design Document
- Firmware Design and Development
- Coding

1.4.2 Testing

- 1.Module Testing (Hardware modules)
- 2.Unit & Integration testing of the software
- 3.System Testing

CHAPTER 2

HARDWARE DESIGN

2.1 BLOCK DIAGRAM

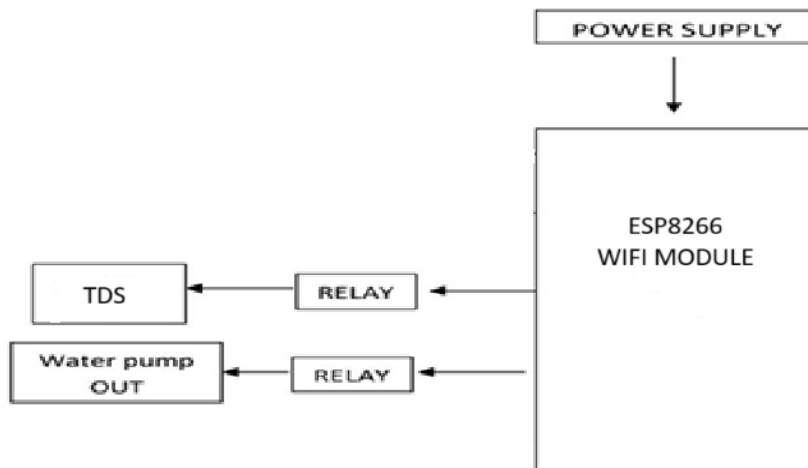


Figure 1:Block Diagram

WORKING:

In every aquaponics system, the first step is to establish a cycle in the system. So, for the formation of bacteria in the grow medium, the fish waste is circulated through the grow bed and from the grow bed the water is filtered and sent back to the tank. In a 7-day period the bacteria required to convert ammonia into the nitrate is formed. Then the plantation is started.

Aquaponics is combination of hydroponics system and aquaculture system. In conventional hydroponics system plants required some mineral fertilizers. In conventional aquaculture system the water required periodical change, by combining these two systems into a bio-controlled looping formation the requirements of addition of mineral fertilizers and exchange of water is satisfied.

Large scale application of aquaponics system has not been reached a successful popularity and the concept is very complex so that people find difficulty to understand the concept of aquaponics like what is happening in aquaponics. Aquaponics can also applied in large scale system but it requires more investment to get an effective aquaponics system. This

projects speaks of explaining the idea of aquaponics in terms of conventional methods of irrigations like canal irrigation. In this project the conventional canal irrigation system is modified for the large scale implementation of aquaponics system. Where all the limitations and applications are considered to implement in large scale system by modifying the conventional irrigation system. For large scale irrigation process the canal irrigation in the form of conventional irrigation can be modified in the form of aquaponics agricultural irrigation system. Where the main canal serve as an aquaculture system and the irrigation fields serves as a hydroponics system. Here the main can has the ability to serve as aquaculture system by receiving large amount of water from canal head work. At the same time the nutrient rich water has been produced by fishes in main canal is discharged to irrigation filed. After the process of uptake by plants the discharged water received by collecting system. Then the water again recirculated into main canal with proper controlling of water quality parameters. In irrigation field racking system also applicable by using this method. Prototype of aquaponics system is made for the observation of behaviours of both plants and fishes.

2.2 ESP8266 WIFI MODULE



Figure 2: ESP8266

The ESP8266 is a low-cost Wi-Fi-enabled microcontroller used for IoT applications and wireless communication. It supports connecting to existing Wi-Fi networks or creating its own hotspot for data transfer. With built-in TCP/IP protocols and

programmable GPIO pins, it can interface with sensors and control devices easily. Its small size, low power consumption, and easy programming make it ideal for smart farming and automation projects.

2.2.1 PIN CONFIGURATION

NODE MCU	GPIO No	Type / function	Description
D0	GPIO16	Digital I/O	No PWM, no interrupts; used for wake-up
D1	GPIO5	Digital I/O, I2C SCL	Commonly used for sensors/motors
D2	GPIO4	Digital I/O, I2C SDA	Used for sensors & modules
D3	GPIO0	Digital I/O	Must be high at boot; avoid using for output
D4	GPIO2	Digital I/O, Built-in LED	Must be high at boot; avoid low at startup
D5	GPIO14	Digital I/O, SPI SCK	Safe for sensors and modules
D6	GPIO12	Digital I/O, SPI MISO	Safe for general use
D7	GPIO13	Digital I/O, SPI MOSI	Safe for general use
D8	GPIO15	Digital I/O, SPI CS	Must be LOW at boot; avoid using for inputs
A0	ADC0	Analog Input	Reads 0–1V input; used for TDS sensor
Vin	-	Power Input	5V supply input
3V3	-	Power Output	3.3V for sensors
GND	-	Ground	Common ground
RST	-	Reset Pin	Resets the module
EN	-	Chip Enable	Must be HIGH for normal operation

Table 1 : ESP8266 PIN CONFIGURATION

2.2.2 FEATURES

- Has built-in Wi-Fi.
- Uses low power.
- Connects easily to sensors.
- Easy to program with Arduino.

2.3 TDS SENSOR

A TDS sensor measures the Total Dissolved Solids in aquaponics water, indicating the levels of nutrients, salts, and minerals available to plants. In a healthy aquaponics system, TDS typically stays between 300–600 ppm, depending on the plant and fish species. Monitoring TDS ensures proper nutrient availability for plants while keeping the water safe for fish. Accurate readings require stable water temperature and converting EC values to TDS using a factor (commonly 0.5 or 0.7). Regular calibration with standard solutions is essential to maintain reliable sensor performance.



Figure 3: TDS SENSOR

2.3.1 TDS Sensor Pin Description

SL NO	PINS	PIN DESCRIPTION
1	A0	Analog output
2	GND	Ground
3	VCC	Power supply(3.3v)

Table 2 : TDS Sensor Pin Description

2.4.2 FEATURES:

- Relays act as automatic switches in aquaponic systems.
- They control pumps using small currents.
- Protect equipment from overload or damage.
- Help run the system safely and efficiently without manual operation.
-

2.5 WATER MOTOR:

The water pump is a key component in aquaponic systems, circulating water between the fish tank and grow beds. It ensures continuous flow, delivering nutrients from fish waste to the plants and maintaining oxygen levels for fish. Controlled by a relay and microcontroller, the pump can operate automatically based on system requirements. Proper pump selection is crucial, considering flow rate, head height, and power efficiency. Overall, it supports a stable and healthy environment for both plants and fish, enabling efficient and sustainable aquaponic farming.



Figure 5: WATER MOTOR

2.5.1 FEATURES:

- Pumps water to circulate between fish tanks and plant beds.
- Provides oxygen-rich water for fish and nutrients for plants.
- Helps maintain water flow and system balance.
- Runs automatically to keep the farm healthy and efficient.

2.6 POWER SUPPLY



Figure 6: 12VDC – 1.5A POWER ADOPTER

2.6.1 Specifications:

- Input Voltage : 100~240V
- Output Power : 12W
- Output Voltage : DC 12V $\pm 5\%$
- Output Current : DC 1.5A
- Frequency : 50/60 Hz
- AC Inlet Mode : 2Pins AC Cord

CHAPTER 3

SOFTWARE DESIGN

3.1 HARDWARE RESOURCE ALLOCATION

The aquaponic system uses several hardware components that together monitor water quality, control feeding, and manage plant irrigation. The allocation of each hardware resource is as follows:

3.1.1 ESP8266 Node MCU (Main Microcontroller + Wi-Fi)

- Acts as the central controller for the entire aquaponic system.
- Performs:
 - Reading TDS, temperature, and moisture values
 - Controlling relays for pumps and feed motor
 - Sending live sensor data to Blynk IoT
- Built-in Wi-Fi eliminates the need for an external IoT module.

3.1.2 Sensors

a. TDS Sensor (Total Dissolved Solids)

- Measures water quality inside the fish tank.
- Detects:
 - Nutrient concentration
 - Fish waste levels
 - Water purity for plant and fish health
- Sends digital/ analog values to ESP8266.

PROGRAMMING LANGUAGE

The programming language used for developing this aquaponic farming system is Embedded C / Arduino C. It is a simplified version of the C/C++ programming language, specially designed for microcontrollers such as the ESP8266, ESP32, and Arduino boards.

Key Features of This Programming Language in Code

1.The language supports Wi-Fi libraries for sending data to Blynk.

- Sensor and Motor Control

2.we can easily read analog values from the TDS sensor and control the water pump using digital pins.

- Real-time automation

3.The code continuously monitors TDS levels and automatically switches the pump ON/OFF.

3.3 DEVELOPMENT TOOLS

The development of the automatic aquaponic system requires both software and hardware tools to design, program, simulate, and test the IoT-based automation. The tools used in this project are:

3.3.1. Arduino IDE

- Primary development environment for writing and uploading code to the ESP8266 Node MCU.
- Supports C/C++ programming.
- Provides library management for TDS, temperature, moisture sensors, Wi-Fi, and Blynk integration.

3.3.2. ESP8266 Board Manager

- Installed within Arduino IDE.
- Allows compiling and uploading code specifically for:
 - Node MCU ESP8266
 - ESP-12E modules
- Required to configure Wi-Fi and Blynk IoT functions.

3.3.3. Blynk IoT Platform

- Used to build the mobile dashboard for:
 - Monitoring TDS values
 - Reading moisture and temperature
 - Controlling water pumps and feed motor
- Provides API commands, virtual pins, and cloud connectivity.

3.3.4. Libraries Used

Several embedded libraries support sensor reading and IoT functions:

- ESP8266WiFi.h – Connects microcontroller to Wi-Fi
- BlynkSimpleEsp8266.h – Cloud data transfer
- TDS Sensor library / Analog processing functions – For TDS measurement
- BlynkTimer – Used for periodic pump running and feeding cycles

3.3.5. Serial Monitor / Serial Plotter

- Tools within Arduino IDE used to:
 - Debug sensor readings
 - Monitor TDS, temperature, and moisture values
 - Check Wi-Fi connection and relay output

3.3.6. Cloud Integration Tools

- Blynk Cloud Dashboard
 - For real-time data logs
 - Remote pump control
 - Push notifications on abnormal TDS/temperature levels
 -

3.4 Design

The design of the automatic aquaponic system integrates hardware, software, and IoT components to create a fully automated plant and fish-growing ecosystem. The system is engineered to continuously monitor water quality, regulate nutrient circulation, and automate feeding cycles while providing real-time data through the internet. The design consists of the following major elements:

3.4.1. System Architecture Design

The system is designed using a recirculating loop that combines aquaculture and hydroponics:

- Fish tank:
Acts as the nutrient source, where fish waste increases TDS and enriches the water.
- Grow Bed Tray:
Plants absorb nutrients and purify the water.
- Water Pumps (IN & OUT):
Maintain circulation between fish tank and plant bed.
- ESP8266 Node MCU:
Serves as the main controller, gathering sensor data and automating all system functions.

3.4.2 Sensor Integration Design

Sensors are placed strategically to monitor critical parameters:

- TDS SENSOR:
Installed in the fish tank to measure dissolved solids indicating water quality

3.4.3. Control System Design

The control system uses the ESP8266 microcontroller to automate mechanical operations:

- Relay Module:
Controls high-voltage Devices (pumps) from the low-voltage ESP8266.
- Pump Operation Design:
Pumps turn ON when:
 - Water circulation timer triggers
 - TDS exceeds safe limits

3.4.4. IoT Connectivity Design

The system includes full IoT integration:

- ESP8266 connects to Wi-Fi.
- Sends real-time data (TDS) to Blynk IoT.
- User can:
 - Monitor water quality anywhere in the world.
 - Turn pumps ON/OFF remotely.
 - Receive alerts for abnormal TDS.
- Ensures fully remote monitoring and control.

3.4.5. Power Supply Design

- ESP8266 powered using 5V USB or regulated adapter.
- Pumps and feed motor powered via 12V or 5V depending on type.
- Relays isolate microcontroller from high-power circuits for safety.

3.4.6. Software Design

The software is structured into modules:

- Sensor reading module
- TDS calculation algorithm
- Decision-making (automation logic)
- Relay control module
- Blynk IoT communication module
- pump timer

This modular design ensures easy debugging and future upgrades.

3.4.7MAIN FLOW DIAGRAM

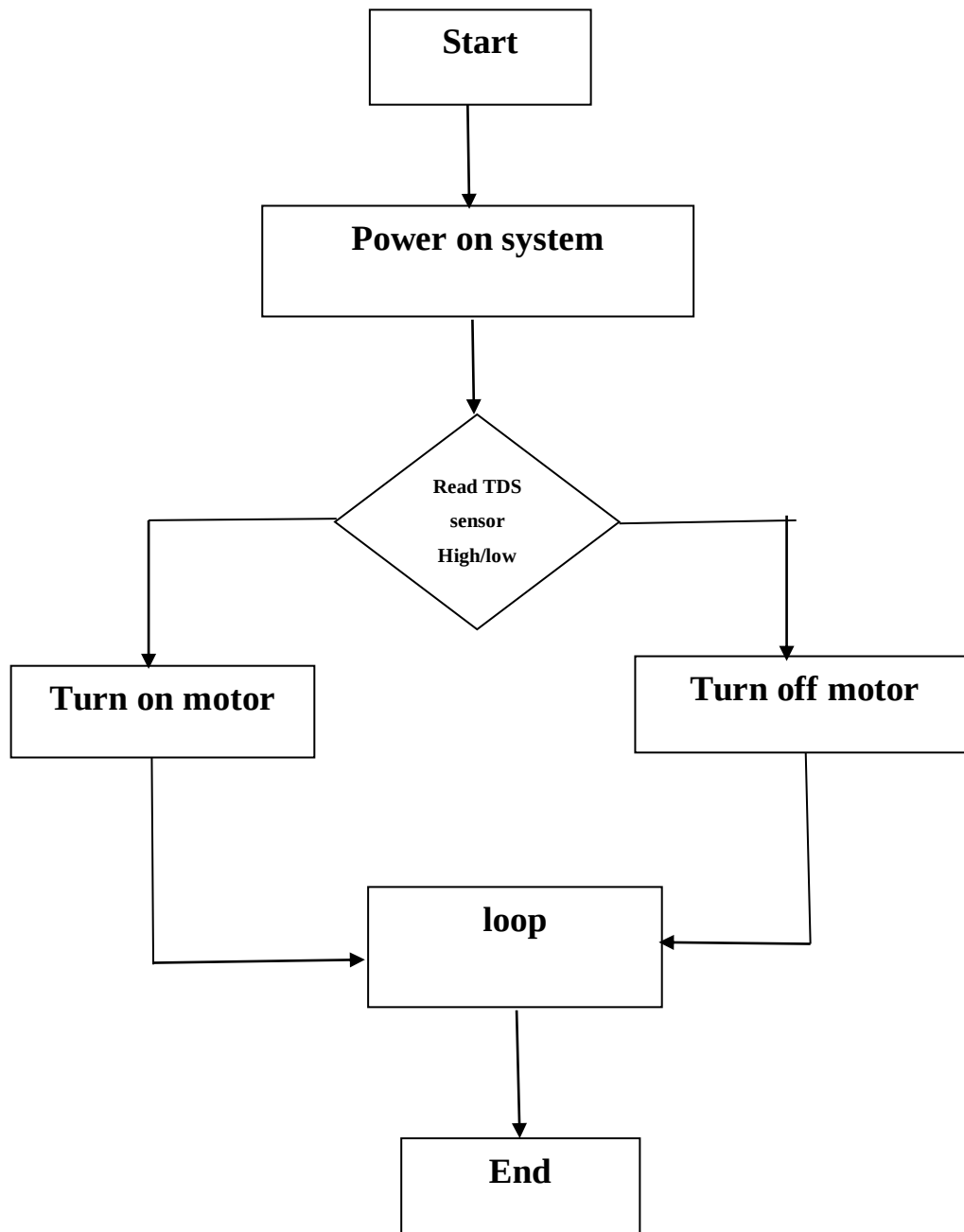


Figure 7:FLOWGRAPH

CHAPTER 4

APPLICATIONS

1. Sustainable Food Production

Aquaponics produces organic vegetables, herbs, and fish without synthetic fertilizers or pesticides. It is widely used for:

- Leafy greens (lettuce, spinach, kale)
- Herbs (basil, mint, coriander)

2. Urban and Rooftop Farming

Aquaponics fits in:

- Small balconies
- Rooftops
- Urban homes
- Apartments

It allows city residents to grow fresh food with minimal space.

3.Home Gardening / Hobby Farming

Many people use aquaponics at home for:

- Growing pesticide-free food
- Hobby gardening
- Backyard systems

8. Disaster Relief & Remote Locations

Aquaponics works well in:

- Places with poor soil
- Drought areas

ADVANTAGES

- 1) Portable
- 2) Environmental friendly.
- 3) Easy to setup
- 4) Light weight.
- 5) Easy maintenance.
- 6) No skill operator required.
- 7) Easy in operation.
- 8) Low cost
- 9) Simple construction.
- 10) Adaptable.
- 11) High capacity.
- 12) Performance.
- 13) Sustainable and intensive food production system.
- 14) Two agricultural products (fish and vegetables) are produced from one nitrogen source. iii. Extremely water efficient.
- 15) Does not require soil. v. Does not use fertilizer and chemical pesticides.
- 16) Higher yields and qualitative production. vii. Higher level of Biosecurity and lower risk from outer contaminants.
- 17) Can be used on non-arable land such as deserts, degraded soil or salty sandy islands.

DISADVANTAGES

- Energy Dependence
- Fish Health Risks
- Limited Crop Variety

CHAPTER 5

RESULT

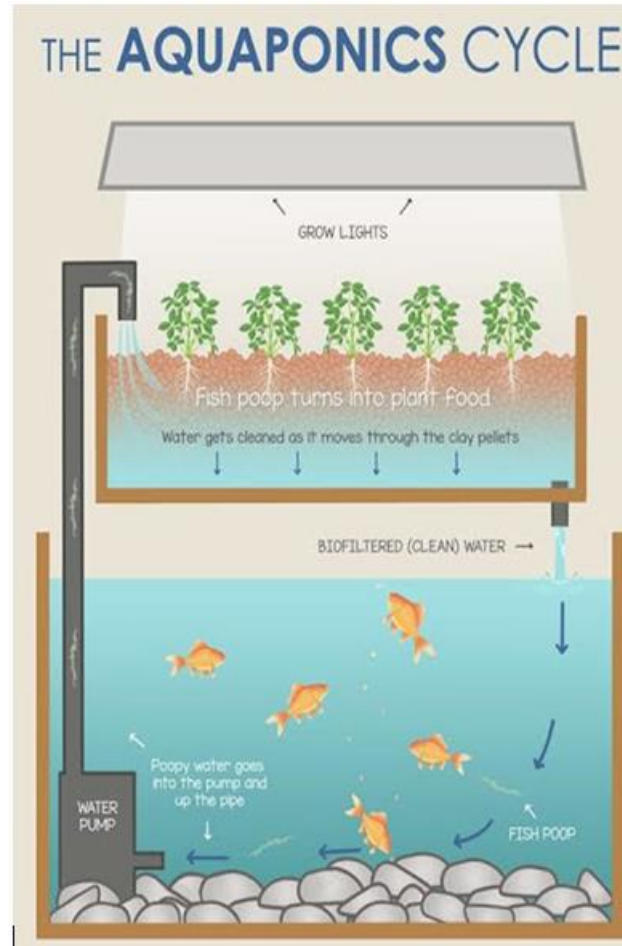


Figure 8:RESULT OF AQUAPONICS FARMING

The aquaponic farming system was successfully developed and tested using a TDS sensor, ESP8266 microcontroller, water motor, relay, and Blynk IoT platform. The system continuously monitored water quality and maintained suitable conditions for plant and fish growth.

The TDS sensor accurately measured the Total Dissolved Solids level in the water. Based on the sensor readings, the water motor operated automatically through the relay to maintain the required nutrient balance. All readings were successfully displayed on the Blynk application in real time.

The system maintained stable TDS values within the recommended range for aquaponics, showing that the automation effectively controlled water quality. The plants showed healthy growth because the nutrient levels were automatically managed, and the fish remained healthy due to clean, balanced water.

Overall, the implementation demonstrated that an automated aquaponic system can reduce manual work, save water, and improve plant growth by maintaining optimal water quality through continuous monitoring and control.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION:

Effective aquaponic farming integrates aquaculture and hydroponics to create a self-sustaining ecosystem. Fish waste provides essential nutrients for plants, while plants naturally filter and purify the water. The TDS (Total Dissolved Solids) sensor monitors water quality, ensuring nutrient levels remain optimal for plant and fish health. Relays control the water pump automatically, maintaining consistent circulation between the fish tank and grow beds. This automation reduces manual labor and prevents system failures due to irregular water flow. Continuous monitoring and pump control promote healthy plant growth and higher crop yields. The system conserves water compared to traditional farming, making it environmentally sustainable. By balancing fish and plant requirements with automated sensors and controls, aquaponics maximizes productivity in limited space. Early detection of water quality issues ensures system reliability and prevents losses. Overall, the integration of sensors, relays, and pumps makes aquaponics an efficient, eco-friendly, and scalable method for sustainable food production.

6.2 FUTURE SCOPE:

OBJECTIVES & SCOPE OF STUDY

- 1) To design a simple structure
- 2) To reduce the human efforts
- 3) To give the human comfort
- 4) To make a device in which we can get the data anywhere in globe
- 5) To grow the plants without soil

CHAPTER 7

REFERENCES

- [1] "Mafrq, Al Mafrq Weather Averages | Monthly Average High And Low Temperature | Average Precipitation And Rainfall Days | World Weather Online". *Worldweatheronline.com* . N.p., 2017. Web. 5 Apr. 2017.
- [2] "Filtration - Pure Hydroponics". *Purehydroponics.com* . N.p., 2017. Web. 5 Apr. 2017.
- [3] "Mafrq, Al Mafrq Weather Averages | Monthly Average High And Low Temperature | Average Precipitation And Rainfall Days | World Weather Online". *Worldweatheronline.com* . N.p., 2017. Web. 5 Apr. 2017.
- [4] Build Your Own Hydroponic Drip System. (n.d.). Retrieved April 09, 2017, from http://www.homehydrosystems.com/system_plans/Drip%20system%20plans/Easy%20to%20Build%20Hydroponic%20Drip%20System/Easy%20to%20Build%20Hydroponic%20Drip%20System.html
- [5] Hydroponic Ebb and Flow, Flood & Drain Systems. (n.d.). Retrieved April 09, 2017, from http://www.homehydrosystems.com/hydroponic-systems/ebb-flow_systems.html
- [6] The Future of Farming: Eight Solutions For a Hungry World. (n.d.). Retrieved April 09, 2017, from <http://www.popsoci.com/environment/article/2009-07/8-farming-solution-help-stop-world-hunger#page-3>